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Basic

ELECTRICITY

COM 223

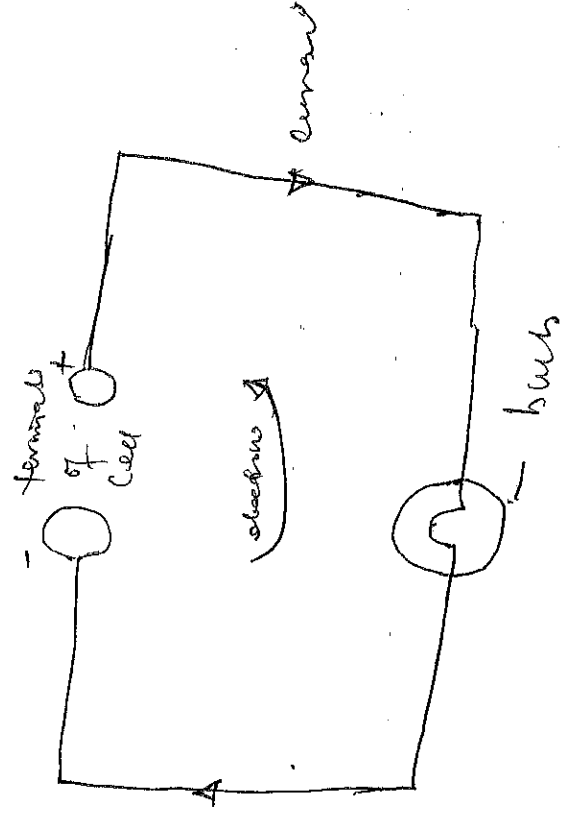
Electrical Circuits

SMITH

(GPT)
style

Electrical current is simply electric charges in motion. In a solid conductor such as a wire consist of a swarm of moving electrons. Electric cells are chemical devices which are capable of causing an electric current to flow. The electrons by producing an electric force which pushes the current along. A current will flow only if there is a complete circuit of conductors by which the current can leave from one end or terminal of the cell and travel round to the other terminal. The current is the same at any point round the circuit and if the line of conductors is broken at any point the current is switched off and stop. One thing needs to be remembered about the electric circuits, we have already said the flow of current is a flow of electrons and these, because they are negatively

charged, flows round the circuit from the negative terminal, or cathode of the cell to the positive terminal or anode. However by convention agreed upon before electron was discovered, we always suppose that an electric current is flow of positive charges from positive terminal to negative terminal. See diagram below.



Types of Electric Cell (Current and Voltage sources)
divided.

Cell may be divided into two main kinds, primary cell and secondary cell.

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Current and Voltage Sources

Primary Cells

Primary cell are those in which current is produce as a result of an irreversible chemical change. The electrolyte and electrode have to be replace when they ~~are~~ ~~them~~ wear out eg the Leclanché cell

Secondary cell

Secondary cell are those which can be recharged when they run down by passing current backward through them. eg the Daniel cell.

Basic ~~the~~ electric quantities.

There are three basic quantities which must be clearly understood before we can ~~proceed~~ proceed with the subject of electricity.

These include

- ① Current (I)
- ② Potential difference (V)
(a) $\frac{W}{Q}$
- (R).

Electric current:

An electric current (I) is the rate of flow of charge round a circuit. The quantity of Charge (Q) is measured in units called Coulombs.

$$\text{Hence Electric current} = \frac{\text{quantity of charge (Coulombs)}}{\text{time (seconds)}}$$

In Symbols

$$I = \frac{Q}{t}$$

The unit of current, one Coulomb per second, is called Ampere (A)

eg. Example. What amount of electricity pass when 18 Coulombs of charge is pass through a wire at a steady state rate of ~~Coulombs~~ in two seconds.

$$I = \frac{Q}{t} = \frac{18}{2} = 9 \text{ A}$$

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Current vary in magnitude from very large values of hundreds of amperes in electric motors to very small amount, microamperes (mA) or nanoamperes (nA) is required for

$$1 \text{ mA} = 10^{-3} \text{ A}$$

$$\text{micro} = 10^{-6} \text{ A}$$

Potential difference (P.d)

The Potential difference (P.d) between any two points in a circuit is the work done when one coulomb of charges moves from one point to the other.

The units of P.d is V.

We have also seen that a cell gives an electric force or pressure across its terminals known as electromotive force (e.m.f).

The e.m.f of cell is used as the P.d between the terminal of the cell when it is not delivering any current to the circuit.

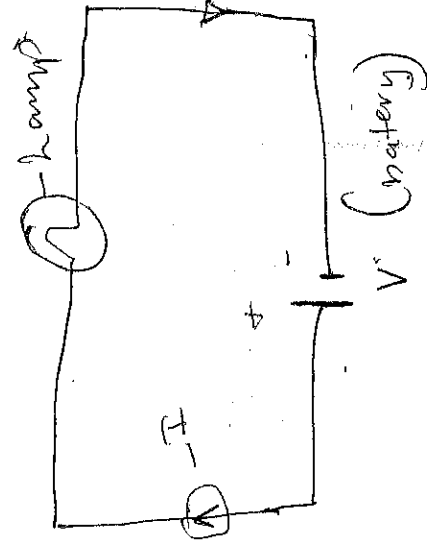
Other units of P_{ind} include

$$1kV = 10^3 V$$

$$1mV = 10^{-3} V$$

The instrument of measuring the voltage is voltmeter and the instrument of measuring current is Ammeter.

The ~~not~~ circuit connection comparing the current (I) and voltage (V).



Resistance

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We have seen different substances have different conducting powers and they are accordingly known as good or poor conductors or insulators.

In electricity we think in terms of the ability of a substance to resist the flow of electricity. A good conductor is therefore said to have a low resistance and a poor conductor has a high resistance. Very good insulators may be regarded as having infinite resistance. The symbol of the ~~Resistance~~ is (R) .

The unit of (R) is Ohm. ~~the~~ (Ω) .

$10^6 \Omega$ Resistor

$10^3 \Omega = 10^3 \Omega$

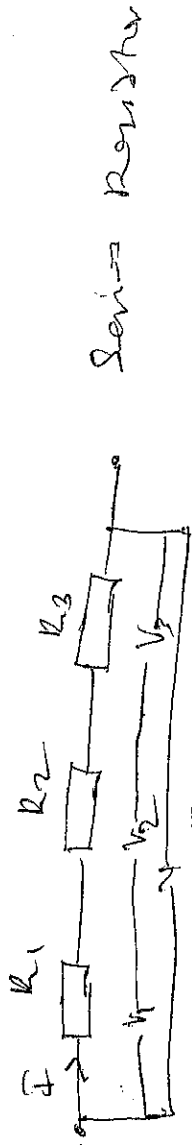
Other elements

(1) Capacitor

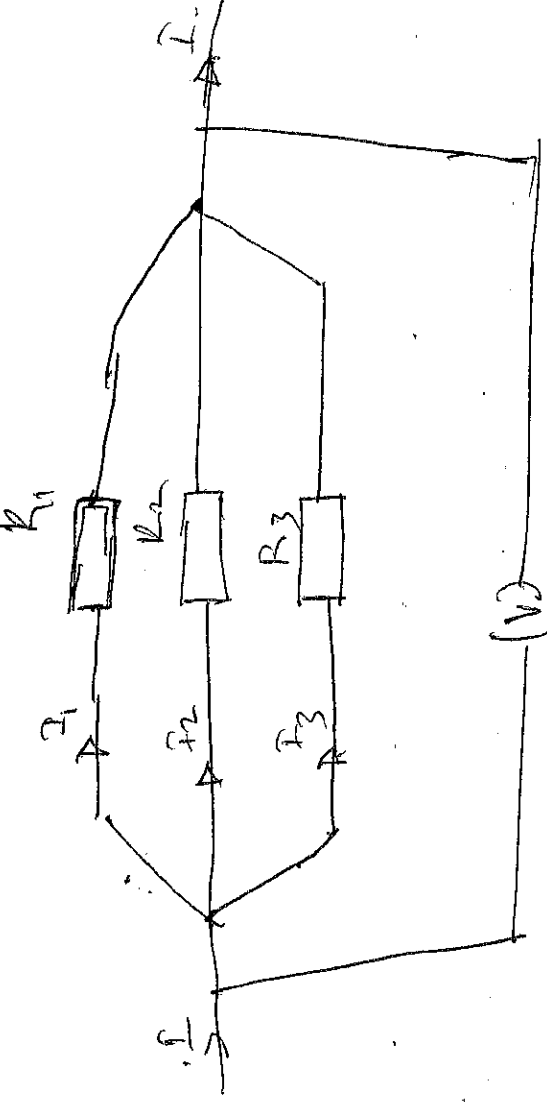
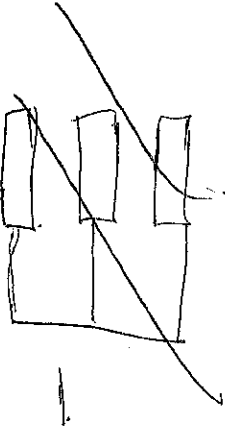
(ii) Inductor

Elements include

Resistors are arranged in either Series or parallel e.g.



Series Resistor



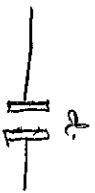
Example what is the parallel combine resistance of 10, 2 and 5.

Sol.

$$\frac{1}{R_T} = \frac{1}{10} + \frac{1}{5} = \frac{3}{10} \Rightarrow R_T = \frac{10}{3}$$

Capacitors

Capacitors are electrical devices that store electrical energy. They are generally in form of two conductors which are insulated electrically from the surroundings. However, most common type of capacitors are in form of two parallel plate conductors which are separated by a distance d . The symbol of capacitor is



Experiment shows that the magnitude of the charge (Q) on any capacitor of the plates is directly proportional to the pd across the capacitor or $Q = CV$ where C is constant of proportionality

The unit of capacitance is the Farad (F). In practice unit are micro (mc) and pico (p) Farad

$$1 \mu F = 10^{-6} F$$

$$1 pF = 10^{-12} F$$

If two or more capacitance $C_1, C_2, \dots, C_n$ are connected in series, the equivalent capacitance C is given by

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n}$$

If connected in parallel then $C = C_1 + C_2 + \dots + C_n$

Inductors

~~Inductors are~~

Inductor are electrical devices that store magnetic energy. They are made of wound coil on a soft iron. The symbol of ~~the~~ inductor is L and is measured in Henry.

Electrical Theories

The Basic electrical Theories include

- (i) Ohm Law
- (ii) Kirchhoff's Voltage Law KVL
- (iii) Kirchhoff's Current Law KCL

① Ohm Law

The current passing through a metallic conductor at a constant temperature is proportional to the potential difference between its ends.

Mathematically

$$V = IR$$

$$R = \frac{V}{I}$$

$$I = \frac{V}{R}$$

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Example 1)

A conductor has a resistance of 2Ω . If a current of 5 ampere (A) is flowing in the wire, what is the pd across the terminals?

$$V = IR = 5 \times 2 = 10 \text{ Volts}$$

KCL

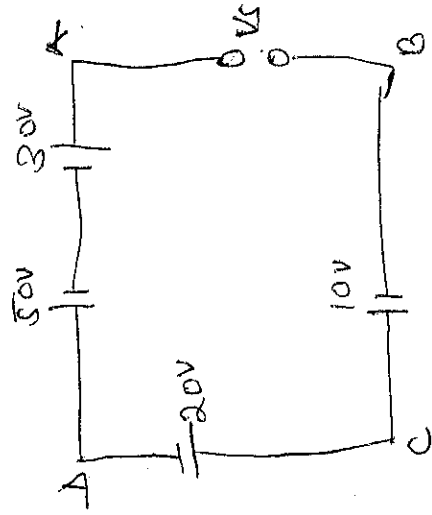
The algebraic sum of the currents entering a junction is equal to the sum of currents leaving a junction. i.e.
Sum of current at a junction is equal to zero
$$I_1 = I_1 + I_2 \quad \text{OR} \quad I - (I_1 + I_2) = 0$$

~~KVL~~

The KVL states that the algebraic sum of the potential differences round any closed path of the circuit is equal to zero.

$$\text{i.e.} \quad \sum (E + IR) = 0$$

Example what is the voltage V_s across the open switch in the circuit below



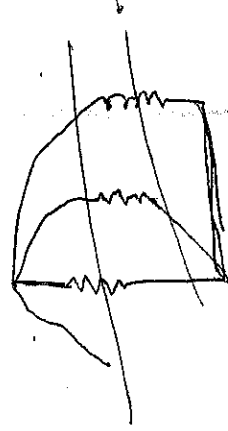
Solution

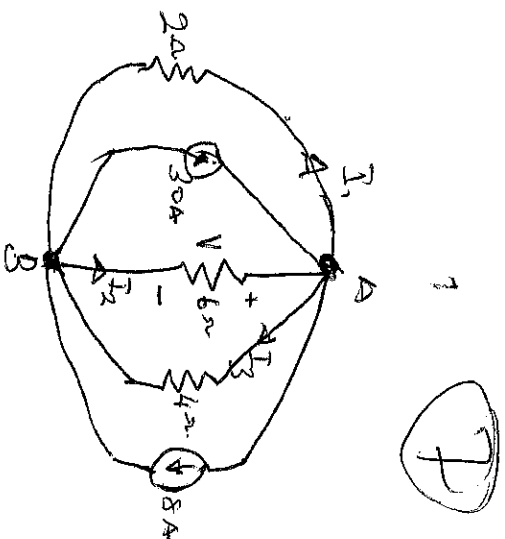
apply KVL to find V_s starting from point A in the clockwise direction, and using the sign convention

$$+V_s + 10 - 20 - 50 + 30 = 0$$

$$V_s = 30V$$

Example ii Using KCL and Ohm law find the magnitude and polarity of voltage V in the figure below.





$$-I_1 + 30 + I_2 - I_3 - 8 = 0 \quad \text{--- (i)}$$

$$\text{or } I_1 - I_2 + I_3 = 22 \quad \text{--- (ii)}$$

but by ohm law

$$I_1 = \frac{V}{2} \quad I_3 = \frac{V}{4} \quad I_2 = -\frac{V}{6} \quad \text{--- (iii)}$$

Substituting equation (iii) in (ii) above we have

$$\frac{V}{2} - \left(-\frac{V}{6}\right) + \frac{V}{4} = 22$$

$$V = 24$$

$$I_1 = \frac{V}{2} = \frac{24}{2} = 12A$$

$$I_2 = -\frac{V}{6} = -\frac{24}{6} = -4A$$

$$I_3 = \frac{V}{4} = \frac{24}{4} = 6A$$

Switches

A Switch is a device that allow you to open and close an electric circuit this lets you turn somethings on or off, or select one of the

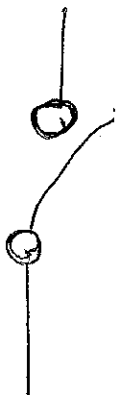
Several choices. Basic electronics component don't get much simpler than a switches, but there are wide assortment of different switches out there, for every possible need.

The most basic switch is two pieces of wire coming together, when you touch the wires together, electricity can pass from one to the other, completing the electric circuit when you pull the wires apart, the circuit is broken.

The simplest of switches is an SPS-T Switch (Single pole Single throw). Imagine a wire connected to the terminal in the middle of the switch. ~~Can make~~ another wire connected to the terminal on the left side the circuit is complete. The 1st switch in your bedroom is a typical example

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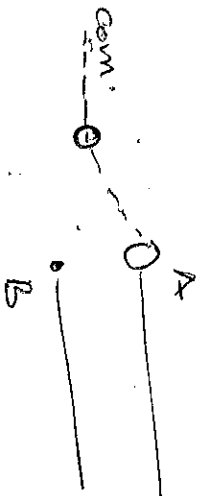
below is the Symbol,



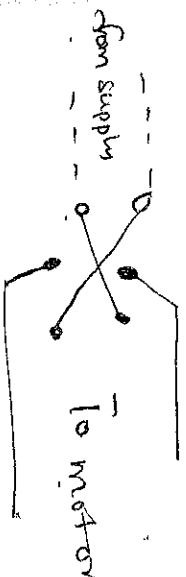
SPST ~~Switch~~ Switch.

Other examples are:

SPDT Single pole Double Throw



DPST - Double pole Single Throw

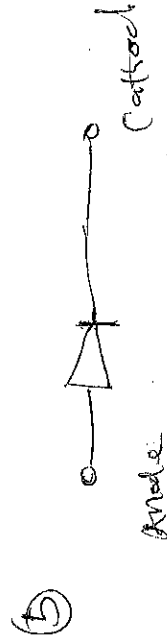


DPDT Double pole Double Throw

Diodes and transistors

Diode

A diode is a two-terminal device consisting of a P-N junction formed either in Ge or Si crystal. Its circuit symbol is shown below. The P and N-type regions are referred to as anode and cathode respectively. An arrow head indicates the direction or conventional direction of flow when forward biased. It is the same direction in which holes flow take place.

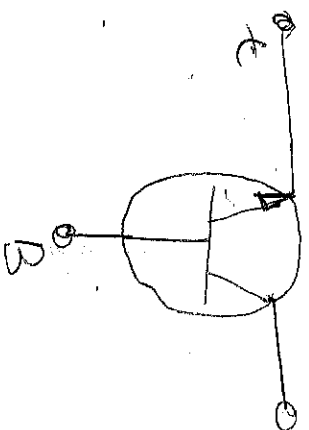
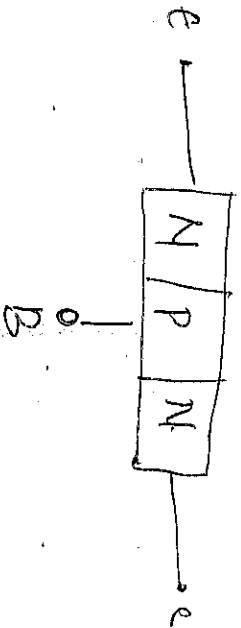
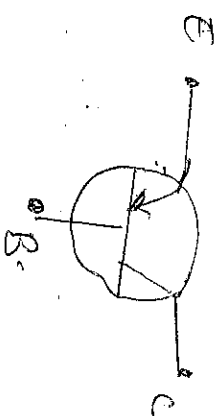
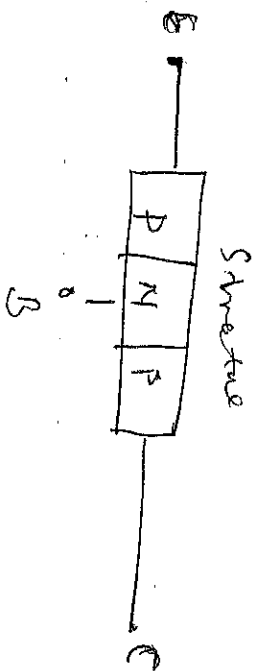


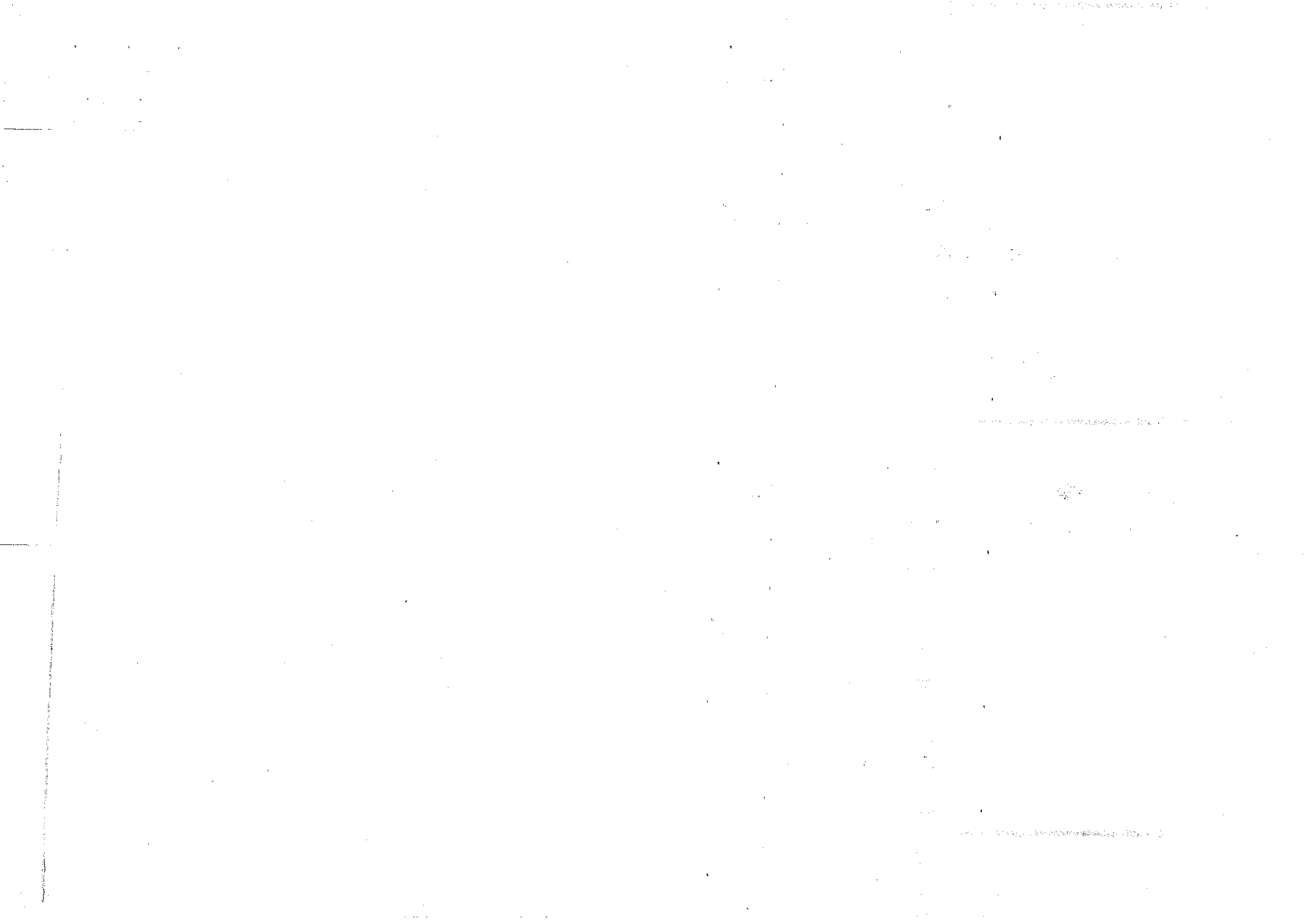
Transistors

Basically the pnp transistor consists of two back to back P-N junctions. It is a single piece of semiconductor. These three regions are called emitter, base, and collector.

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As shown below Junction transistor is a Sandwich of one type of semiconductor material between two layers of the other types. If N-type material Sandwich between two layers of P-type material, it is described as a PNP transistor. If P-type material Sandwich between two layers of N-type material, it is described as an NPN transistor.





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N/D 11 Computer Science

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HARDWARE MAINTA

- ① General Safety in the laboratory
- ② General laboratory etiquette.
- ③ Electrical Safety in Computer Labs
- ④ Safety inspection guide.
- ⑤ Precautions.

Introduction:

A Safety is a condition of being safe from undergoing or causing hurt, injuries or loss. There are also some safety rules laid down by the professional to follow in or to be safe from undergoing your safe machines, and your partner working in the same laboratory!

General Laboratory Safety:

- ① All person working in the laboratory must keep the area clean and orderly
- ② All personal items must be store in cubicle or ...

③ Lab benches must be kept free of extraneous debris at all times.

④ Fumes hoods should not be used to store chemical and should be kept free from non essential items.

⑤ Any Solution or other chemical store in the lab must be properly labelled as to their identity and/or contents.

⑥ The Stockroom or manager should be informed of any chemical store in the lab.

⑦ All dirty glassware is to be placed in the provided bin for collection by the glass cleaner. The contents of difficult to clean glassware must be labelled.

Personal Safety

Rules of the Laboratory

Users of the lab must follow these. Students not following rules will be asked to leave the lab by the instructor or stockroom manager and will not be allowed to return.

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General Laboratory Etiquette.

There can be many hazards when working in labs. But with proper lab techniques and etiquette will reduce the risk of ~~becoming~~ becoming injured. It should be noted that food and drink are not allowed in the lab. Etiquette means behavior below are some of the etiquette.

- ① Always be respectful of your lab mates and their workspace.
- ② Avoid distracting and obstructing lab mates when carrying out an experiment.
- ③ Communicate ~~Questions~~ or discuss around lab safety and proper techniques at all times.

Electrical Safety in Computer Labs.

This is a guide to help you work safer in Electrical and Computer Engineering Laboratories. It provides general guidelines, but does not cover every possible situation specific question regarding lab safety should be addressed to your lab instructor or course instructor. Follow the steps below carefully.

- ① Turn off power and unplug from the wall before working on electric or electronic circuits except when necessary.
- ② Ensure that all naked wire where taken care off. in computer lab.
- ③ Turn off the plugs before turning on your computers and other equipment.
- ④ When you are done with your experiment turn off the power supply.
- ⑤ Do not work on electrical equipment in a wet area always ensure that the ~~an~~

... keep way clean and dry. ③

- ⑥ Turn off power and unplug equipment before checking or replacing fuses.
- ⑦ Immediately replace and don't use defective cords and plugs. Inspect ~~cords~~ cables for defects such as frayed wiring loose connections, or cracked insulation.
- ⑧ Always check the electrical ratings of equipment you use and be sure you used the equipment within its ratings.
- ⑨ Make sure equipment chassis or cabinet grounded. All ~~frays~~ plugs are to be grounded.
- ⑩ Use safe working practices every time.

Safety Inspection Guide

A safety inspection checklist is a document used by employees, supervisors and/or directors to evaluate the work place. This ensures that any potential risks are addressed prior to them occurring and/or avoid potential safety risks.

An inspection checklist is an essential tool for inspectors and professionals across multiple industries and types of business. Using an inspection checklist ensures that the object of inspection goes through a detailed and seen documented evaluation.

What is included in the inspection checklist

- ① Identification of a subject and inspection
 - a) Subject names.
 - b) Location.
 - c) Date of inspection.
 - d) Name and Signature of the inspector.
- ② Inspectors instructions.
- ③ Additional Comment or instruction for the inspectors.
- ④ Checklist of the inspected hazard.
- ⑤ Value of physical quantities e.g. temp. press. & humidity.
- ⑥ Electrical bond. or properties
 - d. Chemical hazard
 - e. Safety protection
 - f. fire protection
 - g. Personal protective equipment
 - h. Noise hazard

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How the inspection check list works:

- ① As each item on the checklist is checked off, the person doing the inspection is verifying that each component of the equipment is in correct working order.
- ② Once the inspection has been completed with the help of a checklist, the objective of inspection can then be endorsed for its intended use.
- ③ Aside from compliance, inspection checklists are also used to validate the safety, quality and/or efficiency of business operations.

Safety rules in Computer Lab

- ① Enter the Computer Lab quietly and move quietly.
- ② Do not Eat or drink while in front of the Computers.
- ③ Do not change the Computer Settings or Backgrounds.

- ④ Be respectful
- ⑤ Clean up your work Area before you leave.
- ⑥ All code and ~~code~~ should be kept under the table headphone should be place under C-pu-tower
- ⑦ Chairs should be push under the table.
- ⑧ All trash paper should be pick off.
- ⑨ Power Down all Computers and monitors before you leave.

Un Safe act

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Unsafe Act

These are the activities or action of the Person, which is possible or certainly lead to accident or increase the risk of accident.

Example :

- (1) Failure to wear your PPE.
- (2) Unsafe driving
- (3) Big Pass or removal of Safety devices
- (4) Using defective equipment
- (5) Use of tools for other than their intended purpose
- (6) Operating a computer/machine which is found to be faulty
- (7) Operating Computer without Competence and training
- (8) Lack of awareness on Safety practice
- (9) Horseplay

